

Mohamed Mostafa

Software Engineer

Banha, Al Qalyubia · mohamed.mostafa0699@gmail.com · +201142314446 · [LinkedIn](#) · [Portfolio](#) · [GitHub](#)

SUMMARY

Software Engineer with 2+ years of experience building production-grade backend systems using Node.js and TypeScript. Proven track record in developing scalable APIs, payment integrations, and monolithic architectures.

EXPERIENCE

Backend Engineer | [Lesoll](#) Jun 2023 – Present · Cairo, Egypt · On-site

- Developed and maintained scalable backend services for the Lesoll platform and Egystay platform.
- Developed key features: Payment system (PayMob & Amazon Payment Services), admin dashboard, chat system, notifications, email/SMS/OTP, traffic scraping (Puppeteer), CoHost system, Cancellation policy, booking, reports, and employee tools.
- Deployed Lesoll via Docker + AWS EC2; deployed Egystay via Docker + AWS EKS + Render.

PROJECTS

Gosha — AI Prompt Optimization Engine

- Architected a self-learning prompt optimization engine that rewrites raw user input into structured, model-specific prompts (Claude XML / GPT Markdown) using a weighted rule system — zero external AI API calls.
- Designed a 5-phase processing pipeline (Analyze → Learn → Transform → Merge → Record) with a feedback loop that adjusts rule weights from user ratings, enabling category-specific personalization over time.
- Built a full SaaS backend with JWT/PASETO auth, tiered subscription billing (Free → Enterprise), dual payment integration (Stripe international + Paymob MENA card & wallet), and a daily token ledger per plan.
- Implemented real-time notifications via SSE backed by Redis pub/sub, async email jobs with BullMQ, structured logging with Pino, and a Prometheus metrics endpoint for observability.

gen-import — TypeScript Barrel File Generator

- Built a zero-dependency CLI tool that auto-generates barrel files (index.ts) for Node.js/Express/TypeScript projects using the TypeScript compiler API for accurate AST-level export detection.
 - Implemented --init command that injects tsconfig path aliases and package.json scripts idempotently, with --alias flag support for custom path prefixes.
 - Added --watch mode using native fs.watch (no chokidar), enabling real-time barrel regeneration on file changes with zero runtime dependencies.
 - Enforced separation of type-only exports (interface, type) from value exports to prevent runtime bootstrap errors; supports CommonJS and ESM output targets.
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EDUCATION

Bachelor of Computer Science · Modern Academy – Maadi, Cairo, Jul 2022

TECHNICAL SKILLS

Languages: HTML, CSS, JavaScript, TypeScript

Backend: Node.js, Express.js, RESTful APIs, Monolithic & Microservices Architecture, Cache, Message Queue, WebSockets

Frontend: Vue.js, Vite, TailwindCSS, UnoCSS, Firebase

Databases: MongoDB, PostgreSQL — Design & Optimization, Transactions, Indexing

Cloud & DevOps: AWS (EC2, EKS), Docker & Docker Compose, CI/CD (GitHub Actions), Nginx

Tooling: BullMQ, Pino, Prometheus, Socket.IO, SSE, Redis, Puppeteer, TypeScript Compiler API